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**General Comments :** Consolidated HW02 submission document uniting the UP Inception Overview (Vision, Personas, Requirements, Business Rules, Use Cases, Glossary) and the Agile Scrum Product Backlog, Themes/Epics, User Stories, Tasks, Sprint Plan, and PSI/MVP definition for the CSE305 Library Management System.

# HW02 — UP Inception Overview & Agile Scrum Product Backlog

## Library Management System (LMS)

| | |
|---|---|
| **Course** | CSE305 Software Engineering — Adnan Menderes University (ADU), Department of Computer Engineering |
| **Instructor** | Assoc. Prof. Dr. Fatih Soygazi |
| **Project** | Library Management System (LMS) |
| **Team & Scrum roles** | **Ibrahim Temur** ( `231805033` ) — Product Owner · **Omer Faruk Kuş** ( `231805026` ) — Scrum Master (both: Developer, Architect, QA, Reviewer) |
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| **Archive** | `cse305_hw02_231805033_231805026.zip` (created with `python3 tools/package.py 02` ) |
| **GitHub repository** | `https://github.com/ibrahimtemur/cse305-library-management-system` |
| **Trello sprint board** | `https://trello.com/b/34K6O3Y1/cse305-lms-sprint-board` |

> **Note**
> **How to read this document.** Part 1 provides a self-contained executive consolidation of the Unified Process (UP) Inception artifacts and links to the authoritative detail files in `docs/02-inception/` : `docs/02-inception/VISION_AND_CONTEXT.md` , `docs/02-inception/PERSONAS.md` , `docs/02-inception/REQUIREMENTS.md` ( `FR-001` .. `FR-032` , `NFR-001` .. `NFR-012` , `CON-001` .. `CON-006` , `DR-001` .. `DR-007` ), `docs/02-inception/BUSINESS_RULES.md` ( `BR-001` .. `BR-018` ), `docs/02-inception/USE_CASES.md` ( `UC-01` .. `UC-11` ), and `docs/02-inception/DOMAIN_GLOSSARY.md` . Part 2 presents the complete Agile Scrum deliverables: Themes & Epics ( `docs/02-inception/THEMES_EPICS.md` ), User Stories ( `docs/02-inception/USER_STORIES.md` , `LMS-001` .. `LMS-021` ), Ordered Product Backlog ( `docs/02-inception/PRODUCT_BACKLOG.md` ), Three-Sprint Iteration Plan ( `docs/02-inception/SPRINT_PLAN.md` ), Task Breakdown ( `docs/02-inception/TASKS.md` , `105` tasks), and Potentially Shippable Increment / MVP Definition ( `docs/02-inception/PSI_AND_MVP.md` ). Every story point, classic hour, human hour, capacity calculation, commitment line, and traceability check is verified deterministically by `tools/agile/validate_backlog.py` .

**Contents:** Part 1 — UP Inception Overview · Part 2 — Agile Scrum Product Backlog & Iteration Plan · Revision History

# Part 1 — UP Inception Overview

## 1.1 Vision, Problem Recap, Goals & System Context

Full specification: `docs/02-inception/VISION_AND_CONTEXT.md` .

### Product Vision Statement

**For** university, faculty, and departmental libraries that manage physical book collections for academic staff, administrative staff, and students, **who** struggle with error-prone manual ledgers, untracked physical book copies, inconsistent overdue penalty calculations, and lack of real-time shelf visibility, **the Library Management System (LMS)** is a self-contained, web-based library catalog and circulation management application **that** automates bibliographic cataloging, barcode-level physical copy lifecycle tracking ( `AVAILABLE` , `LOANED` , `RESERVED` , `MAINTENANCE` , `LOST` ), policy-driven lending and renewal rules, First-In-First-Out (FIFO) title reservation queues, and deterministic overdue fine calculation. **Unlike** informal spreadsheet tracking or heavyweight commercial integrated library systems requiring external institutional infrastructure, **our product** operates 100% offline out-of-the-box with an embedded H2 database seeded from public-domain Open Library metadata ( `782` books, `1,706` physical copies) and strictly synthetic operational data ( `60` members, `3` librarians, `164` loans, `47` fines, `22` reservations), enforcing the confirmed Adnan Menderes University (ADU) library lending rules through modular GoF/GRASP design patterns and a JUnit 5 Test-Driven Development (TDD) test suite.

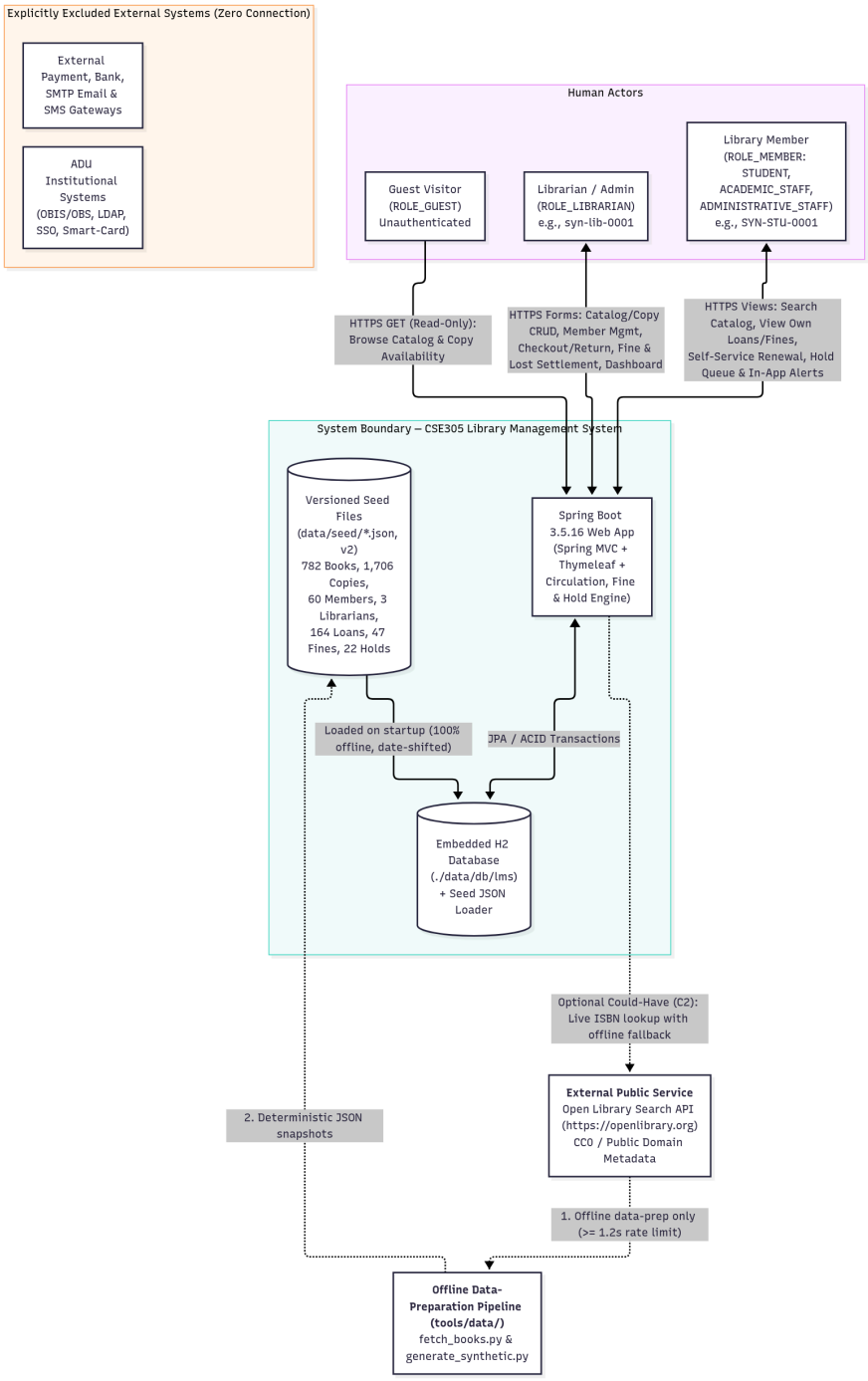
### Problem Recap & Project Goals ( `GOAL-01` .. `GOAL-05` )

Small-to-medium academic libraries using paper logs or unvalidated spreadsheets face four bottlenecks: (1) clerical overhead and manual date/ID errors at checkout/return, (2) inability to distinguish shelf-available copies from loaned, reserved, maintenance, or lost copies (or non-circulating reference works), (3) inconsistent enforcement of member quotas ( `7` / `3` / `3` ), `0.25 TRY/day` fines, post-renewal cooldowns, and borrowing blocks, and (4) zero self-service visibility for members.

| Goal / Metric ID | Category | Summary Statement & Verification Basis |
|---|---|---|
| `GOAL-01` / `MET-01` | **Operational Automation & Policy Accuracy** | Replace manual ledgers with transactional checkout, return, renewal, reservation, and fine settlement governed by `BR-001` .. `BR-018` . **Target ( `ASSUMPTION` ):** `100%` pass rate across JUnit 5 tests covering quotas ( `7` / `3` / `3` ), `30` -day periods, `0.25 TRY/day` uncapped fines, `30` -day post-renewal cooldown, and blocks. |
| `GOAL-02` / `MET-02` | **Copy-Level Inventory & State Integrity** | Track every physical `BookCopy` ( `LMS-COPY-1001` ..) via `BookCopyState` and block checkout of `referenceOnly` books ( `BR-014` , `BR-017` ). **Target ( `ASSUMPTION` ):** `0` illegal transitions or double-loaned copies; `100%` 1-to-1 invariant between `LOANED` copies and `ACTIVE` loans, and between `RESERVED` copies and `READY` holds. |
| `GOAL-03` / `MET-03` | **Self-Service & Local Query Latency** | Empower members to search the catalog, inspect loans/fines/cooldowns, renew, and place/cancel holds ( `UC-01` , `UC-07` , `UC-08` , `UC-10` ). **Target ( `ASSUMPTION` ):** `< 200 ms` local server response time on the reference workstation against the seeded H2 dataset ( `NFR-005` ). |
| `GOAL-04` / `MET-04` | **CSE305 Engineering Rigor & Startup Speed** | Apply UP modeling, Agile Scrum, UI-First prototyping via **Stitch MCP** ( `NFR-001` ), GoF/GRASP patterns ( `NFR-010` ), and TDD ( `NFR-011` ). **Target ( `ASSUMPTION` ):** `< 10 seconds` offline startup and seed initialization time with zero runtime internet calls ( `NFR-004` , `NFR-006` ). |
| `GOAL-05` / `MET-05` | **Offline Reproducibility & Synthetic Privacy** | `0` real personal names, student IDs, emails, or plaintext secrets in Git history; `100%` of seeded emails use `@synthetic.lms.local` and demo passwords are BCrypt-hashed at startup ( `NFR-008` , `NFR-009` , `BR-018` ). |

### System Scope & Explicit Exclusions

- **In Scope:** Printed books only (including non-circulating `referenceOnly` encyclopedias, dictionaries, and catalogs, `BR-014` ); copy-level inventory ( `1,706` seeded copies); synthetic `STUDENT` , `ACADEMIC_STAFF` , and `ADMINISTRATIVE_STAFF` members; desk checkout/return, self-service/desk renewal, `0.25 TRY/day` uncapped fine calculation, desk payment/waiver, lost-copy petition ( `lostReportedOn` , fine freeze, 30-day replacement window, `2x` runtime `appraisedValue` ), FIFO title reservation queue with 3-day pickup hold and in-app notification, librarian KPI dashboard, and Stretch enhancements ( `ROLE_GUEST` portal, CSV export, live Open Library ISBN lookup helper).
- **Explicitly Out of Scope ( `CON-002` .. `CON-004` ):** (1) **Periodicals/serials and unpublished theses** (although ADU regulations allow borrowing 2 non-latest periodical issues for 2 days and restrict unpublished theses, LMS manages books only); (2) **Live ADU OBIS/OBS, LDAP/SSO, or RFID/smart-card hardware** (personal ID card verification `BR-016` is performed by the Librarian at the desk using `memberId` ); (3) **External payment/banking APIs and external SMTP email/SMS gateways** (fines are settled at the desk as `PAID` / `WAIVED` ; hold-ready alerts are rendered in-app).




## 1.2 Stakeholders & User Personas

Full specification and circulation scenarios: `docs/02-inception/PERSONAS.md`. Every persona is anchored to real synthetic records in `data/seed/*.json`.

| # | Persona & Role | Member Type (`memberType`) | Representative Seed Record (`data/seed/*.json`) | Confirmed Quota (BR-002) | Confirmed Period (BR-003) | Core Operational / Self-Service Goals |
|---|---|---|---|---|---|---|
| 1 | **Librarian / Admin** (`ROLE_LIBRARIAN`) | Staff Account (ASSUMPTION-01) | `syn-lib-0001` (Enginiz Yıldırım) in `librarians.json` *(colleagues: `syn-lib-0002`, `syn-lib-0003`)* | N/A | N/A | Fast desk checkout/return by `memberId` & `barcode`, catalog/copy CRUD, member management, fine settlement/waiver (`settledBy`), lost-copy assessment, KPI dashboard (`UC-03`..`UC-07`, `UC-09`, `UC-11`). |

| # | Persona & Role | Member Type (`memberType`) | Representative Seed Record (`data/seed/*.json`) | Confirmed Quota (`BR-002`) | Confirmed Period (`BR-003`) | Core Operational / Self-Service Goals |
|---|---|---|---|---|---|---|
| **2** | **Student Member** (`ROLE_MEMBER`) | `STUDENT` | `SYN-STU-0001` (`Tutkucan Ergül`, Chemistry, `LOAN-0099`) & `SYN-STU-0003` (`Hazrat Yorulmaz`, Computer Eng., `RES-0009` `READY` on `LMS-COPY-1599`) | **3 books** | **30 days** | Search course textbooks, check availability & `referenceOnly` badges, view own active/historical loans, cooldowns & fines, renew once online, join FIFO hold queue (`UC-01`, `UC-07`, `UC-08`, `UC-10`). |
| **3** | **Academic Staff Member** (`ROLE_MEMBER`) | `ACADEMIC_STAFF` | `SYN-STF-0003` (`Atasagun Öcalan`, Computer Eng., `RES-0007`) & `SYN-STF-0002` (`Baydu Arsoy`, Mathematics, `RES-0001`) | **7 books** | **30 days** | Borrow up to `7` research titles simultaneously (`AcademicStaffLoanPolicy`), inspect copy counts for course reading lists, renew non-reserved titles, reserve high-demand literature (`UC-01`, `UC-07`, `UC-08`, `UC-10`). |
| **4** | **Administrative Staff Member** (`ROLE_MEMBER`) | `ADMINISTRATIVE_STAFF` | `SYN-ADM-0001` (`Doğannur Güçlü`, Business Admin.) & `SYN-ADM-0004` (`Peren Erdoğan`, Computer Eng.) | **3 books** | **30 days** | Browse general & professional books, borrow up to `3` books in person with university staff ID card (`AdministrativeStaffLoanPolicy`), track due dates and renew online (`UC-01`, `UC-07`, `UC-10`). |
| **5** | **Guest Visitor** (`ROLE_GUEST`) | Unauthenticated | Anonymous browser session | **0 books** | N/A | Browse and filter the public catalog and inspect shelf availability without logging in, with zero access to member or loan data (`UC-01`, `FR-030`). |

## 1.3 Supplementary Specification Overview (`FR`, `NFR`, `CON`, `DR`)

Full specification of all `32` Functional Requirements, `12` Non-Functional Requirements, `6` Constraints, and `7` Data Requirements: `docs/02-inception/REQUIREMENTS.md` and `data/README.md`.

### Summary of Requirement Categories & Seed v2 Counts

| Category | ID Range | Count | Breakdown & Scope Summary |
|---|---|---|---|
| **Functional Requirements** | `FR-001` .. `FR-032` | **32** | **19 Must Have** (`FR-001` .. `FR-019` across `M1` – `M6`), **10 Should Have** (`FR-020` .. `FR-029` across `S1` – `S4`), **3 Could Have** (`FR-030` .. `FR-032` across `C1` – `C3`). |
| **Non-Functional Requirements** | `NFR-001` .. `NFR-012` | **12** | Usability (`NFR-001` Stitch MCP UI-first, `NFR-002` desk barcode efficiency), Reliability (`NFR-003` `@Transactional` ACID, `NFR-004` 100% offline runtime), Performance targets (`NFR-005` `< 200 ms`, `NFR-006` `< 10 s` startup), Security/Privacy (`NFR-007` RBAC/validation, `NFR-008` BCrypt/no secrets, `NFR-009` synthetic PII), Maintainability/Testability/Portability (`NFR-010` .. `NFR-012`). |
| **System Constraints** | `CON-001` .. `CON-006` | **6** | English & mandatory header policy (`CON-001`), books-only scope (`CON-002`), no live ADU integration (`CON-003`), no payment/SMTP/SMS gateways (`CON-004`), `tools/set-author` shared-PC rule (`CON-005`), protected `main` & dual ADUZEM upload (`CON-006`). |
| **Seed Data Requirements** | `DR-001` .. `DR-007` | **7** | `books-openlibrary.json` (**`782` books**: `779` circulating, `3` `referenceOnly`), `copies.json` (**`1,706` copies**: `1,624` `AVAILABLE`, `59` `LOANED`, `15` `MAINTENANCE`, `5` `LOST`, `3` `RESERVED`), `members.json` (**`60` members**: `40` `STUDENT`, `10` `ACADEMIC_STAFF`, `10` `ADMINISTRATIVE_STAFF`; `55` `ACTIVE`, `5` `SUSPENDED`), `librarians.json` (**`3` librarians**), `loans.json` (**`164` loans**: `59` `ACTIVE`, `100` `RETURNED`, `5` `LOST`), `fines.json` (**`47` fines**: `26` `UNPAID` totaling `187.50 TRY`, `16` `PAID`, `5` `WAIVED`), `reservations.json` (**`22` reservations**: `19` `WAITING`, `3` `READY`). |

### Highlights of the Most Critical Functional & Non-Functional Requirements

| ID(s) | Priority / Category | Condensed Requirement Statement | Linked `BR-*` / `UC-*` |
|---|---|---|---|
| **`FR-001` .. `FR-005`** | Must (`M1` Catalog & Copies) | Librarian CRUD on `Book` (`isbn13` checksum, 11 categories, `referenceOnly` flag for encyclopedias/dictionaries/catalogs), public/member search & detail view with `REFERENCE ONLY` badge, barcode registration (`LMS-COPY-*`), and GoF **State** machine (`BookCopyState`) over `AVAILABLE`, `LOANED`, `RESERVED`, `MAINTENANCE`, `LOST`. | `BR-001`, `BR-014`, `BR-015`, `BR-017` · `UC-01`, `UC-03` |

| ID(s) | Priority / Category | Condensed Requirement Statement | Linked BR-* / UC-* |
|---|---|---|---|
| **FR-009 .. FR-013** | Must ( M3 Checkout & Return) | Atomic @Transactional desk checkout verifying personal ID ( memberId , no proxy), ACTIVE status, quota ( 7 / 3 / 3 ), zero overdue loans, zero UNPAID fines, 30-day post-renewal cooldown on isbn13 , and copy eligibility ( !referenceOnly , AVAILABLE or matching READY hold); atomic return computing overdue fines and setting copy state. | BR-002 , BR-003 , BR-006 .. BR-008 , BR-011 , BR-013 , BR-014 , BR-016 , BR-017 · UC-05 , UC-06 |
| **FR-014 , FR-015** | Must ( M4 Overdue & Fines) | Zero-grace-period overdue calculation overdueDays = max(0, (effectiveEndDate - dueDate).days) (using lostReportedOn , returnDate , or currentDate ) and uncapped daily fine amount = 0.25 * overdueDays TRY via FineCalculationStrategy . | BR-007 .. BR-010 , BR-015 · UC-06 , UC-09 , UC-11 |
| **FR-016 .. FR-019** | Must ( M5 Auth/Self-Service & M6 Seed Loader) | Local login/logout with startup-seeded BCrypt demo passwords ( OPEN-05 ), endpoint RBAC ( ROLE_LIBRARIAN , ROLE_MEMBER , ROLE_GUEST ), member self-service portal + librarian inspection ( FR-008 , FR-018 ), and startup JSON seed loader with date shifting ( delta = today - 2026-10-22 , OPEN-04 ). | BR-001 , BR-002 , BR-006 , BR-010 , BR-013 , BR-017 , BR-018 · UC-02 , UC-04 , UC-10 |
| **FR-020 .. FR-023** | Should ( S1 FIFO Hold Queue) | Title-level reservation when 0 copies are AVAILABLE and !referenceOnly ; strict FIFO queuePosition = 1..n ; GoF **Observer** hold promotion on return ( copy -> RESERVED , reservation -> READY , holdUntil = readyOn + 3 days , in-app notification); cancellation & 3-day expiry reallocation. | BR-011 .. BR-014 · UC-06 , UC-08 , UC-10 |
| **FR-024 .. FR-029** | Should ( S2 Renewal, S3 Fine/Lost, S4 Dashboard) | Single +30 -day renewal if not overdue and unreserved ( FR-024 ); desk fine PAID and WAIVED settlement ( FR-025 , FR-026 ); lost-copy petition ( lostReportedOn freezes fine, 30-day window, 2.00 * appraisedValue , FR-027 ); dashboard KPI cards ( FR-028 ) and overdue/unpaid report ( FR-029 ). | BR-003 .. BR-005 , BR-007 .. BR-011 , BR-015 , BR-017 · UC-07 , UC-09 , UC-11 |
| **NFR-001 , 003 , 004 , 008 .. 011** | Non-Functional Core | UI-first Stitch MCP prototypes in docs/03-design/ui/ before Thymeleaf views; @Transactional ACID integrity; 100% offline build & test execution; zero committed passwords/PII; 5-tier layered architecture with GoF/GRASP patterns and JUnit 5 + Python unittest suites. | Architecture & QA gates |

## 1.4 Business Rules Overview ( BR-001 .. BR-018 )

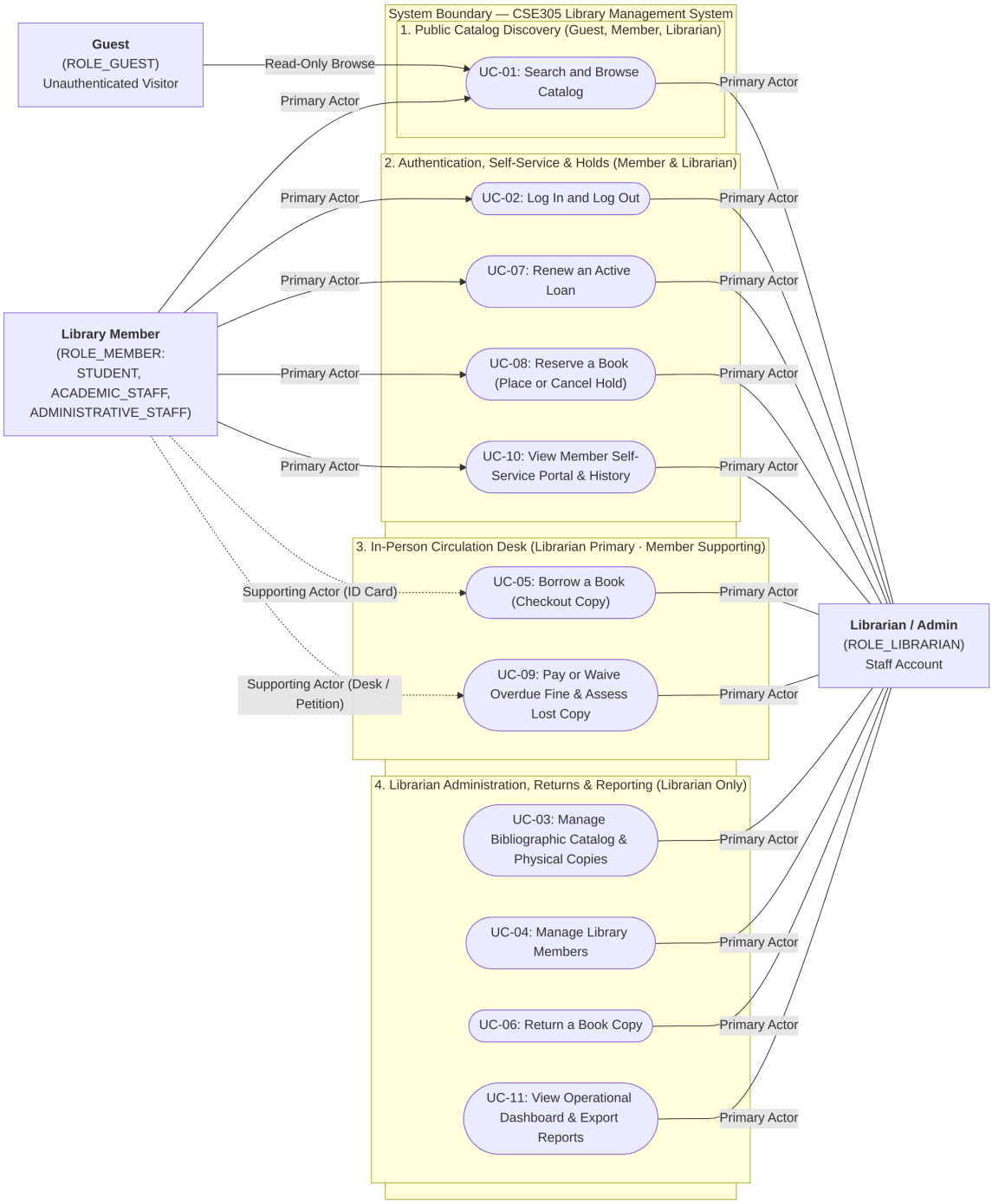
Full catalog of all 18 business rules, official ADU source citations (verified 2026-10-22 ), HW01 assumption disposition table ( ASSUMPTION-01 .. 09 ), new assumptions ( ASSUMPTION-10 .. 17 ), and exact BigDecimal / LocalDate formulas: docs/02-inception/BUSINESS_RULES.md .

| Rule ID(s) | Rule Name | Summary of Rule & Deterministic Formula | Verification Status |
|---|---|---|---|
| **BR-002 , BR-003** | **Borrowing Quotas & 30-Day Loan Period** | Max simultaneous ACTIVE loans: **ACADEMIC_STAFF = 7** , **ADMINISTRATIVE_STAFF = 3** , **STUDENT = 3** . Standard loan duration is **30 calendar days** for all three member types ( dueDate = checkoutDate + 30 days ). | **CONFIRMED** *(ADU Official Rules, 2026-10-22)* |
| **BR-004 , BR-005** | **Single Renewal & Renewal Prohibitions** | An ACTIVE loan may be extended at most **once** ( renewalCount: 0 -> 1 , dueDate = checkoutDate + 60 days ), and **cannot** be renewed if currentDate > dueDate (overdue) or if any WAITING / READY reservation exists on isbn13 . | **CONFIRMED** ( BR-004 ) + ASSUMPTION-11 ( BR-005 : max 1 renewal of +30 days) |
| **BR-006** | **Post-Renewal 1-Month Re-Borrowing Cooldown** | When a renewed loan ( renewalCount >= 1 ) is returned on returnDate , the same memberId cannot borrow any copy of the same isbn13 until returnDate + 30 calendar days ( currentDate < returnDate + 30 days is rejected). | **CONFIRMED** *(1-month rule)* + ASSUMPTION-12 *(30-day mapping)* |
| **BR-007 .. BR-010** | **Overdue Days, 0.25 TRY/day Tariff (No Cap) & Settlement** | Zero grace period: overdueDays = max(0, (effectiveEndDate - dueDate).days) . Daily fine is **0.25 TRY (25 kuruş) per overdue day** with **no maximum cap** ( amount = 0.25 * overdueDays ). At most one Fine per loanId ; settled as PAID or WAIVED by a Librarian ( settledOn , settledBy ) after loan resolution. | **CONFIRMED** ( 0.25 TRY/day ) + ASSUMPTION-10 *(no cap; 250 TRY cap removed)* |
| **BR-011** | **Borrowing & Renewal Block** | Any member holding **>= 1 overdue active loan** OR **>= 1 UNPAID fine ( amount > 0.00 TRY )** OR status == SUSPENDED is blocked from borrowing new books or renewing existing loans. | **CONFIRMED** *(overdue & unpaid fine block)* + ASSUMPTION-06 *( SUSPENDED )* |
| **BR-012 , BR-013** | **FIFO Reservation Queue & 3-Day Pickup Hold** | Title-level hold ( isbn13 ) allowed only when 0 copies are AVAILABLE (and member is ACTIVE , not already borrowing/queuing for isbn13 ). On copy return, copy -> RESERVED , queuePosition == 1 reservation -> READY ( readyCopyBarcode , readyOn = returnDate , holdUntil = readyOn + 3 days ), and an **in-app notification** is created. | **CONFIRMED** *(notification & 2–3 day hold)* + ASSUMPTION-13 *(FIFO, in-app, 3 days)* |
| **BR-014** | **Non-Circulating Reference Works ( referenceOnly )** | **Encyclopedias, dictionaries, and catalogs** have referenceOnly = true and cannot be borrowed or reserved. *(Periodicals and unpublished theses are Out of Scope per CON-002 ).* | **CONFIRMED** *(ADU Official Rules)* + ASSUMPTION-14 *( referenceOnly boolean)* |

| Rule ID(s) | Rule Name | Summary of Rule & Deterministic Formula | Verification Status |
|---|---|---|---|
| `BR-015` | **Lost-Copy Petition, Fine Freeze & 2x Valuation Penalty** | Filing a lost-book petition ( `lostReportedOn` ) **immediately freezes daily overdue fine accrual on `lostReportedOn`** and opens a 30-day replacement window. If unreplaced after 30 days, `copy -> LOST` , `loan -> LOST` , and the Librarian enters the Valuation Commission's `appraisedValue` at runtime to assess **`compensationAmount = 2.00 * appraisedValue`** plus frozen overdue fine. | **`CONFIRMED`** *(freeze, 1-month window, 2x valuation)* + `ASSUMPTION-12` , `ASSUMPTION-15` |
| `BR-016` .. `BR-018` | **Personal ID Card, Copy State Invariant & Synthetic Auth** | Borrowers must present their own personal university ID card ( `memberId` ; **no proxy borrowing**, `BR-016` ). Every copy has a single valid state ( `BR-017` ). Seed JSON contains 100% synthetic PII and zero passwords; BCrypt demo passwords are initialized at startup ( `BR-018` ). | **`CONFIRMED`** ( `BR-016` ) + `ASSUMPTION-08` , `09` , `16` , `17` |

## 1.5 Use Case Model Overview ( `UC-01` .. `UC-11` )

Full specifications (including the **4 fully-dressed** use cases `UC-01` , `UC-05` , `UC-06` , `UC-09` with preconditions, postconditions, main success scenarios, and alternative/exception flows, plus the **7 brief** use cases): `docs/02-inception/USE_CASES.md` .



| UC ID | Use Case Name | Primary Actor(s) | Detail Level in USE_CASES.md | Realized FR-* IDs | Enforced BR-* IDs | Implementing Stories |
|---|---|---|---|---|---|---|
| UC-01 | **Search and Browse Catalog** | Guest, Member, Librarian | **Fully-Dressed** | FR-002, FR-003, FR-030 | BR-001, BR-014, BR-017 | LMS-004 (S1), LMS-018 (S3) |
| UC-02 | **Log In and Log Out** | Member, Librarian | Brief | FR-016, FR-017 | BR-001, BR-018 | LMS-003 (S1) |
| UC-03 | **Manage Bibliographic Catalog & Physical Copies** | Librarian | Brief | FR-001, FR-004, FR-005, FR-031 | BR-001, BR-014, BR-017 | LMS-005 (S1), LMS-020 (S3) |
| UC-04 | **Manage Library Members** | Librarian | Brief | FR-006, FR-007, FR-008 | BR-001, BR-002, BR-011, BR-018 | LMS-006 (S2), LMS-013 (S2) |
| UC-05 | **Borrow a Book (Checkout Copy)** | Librarian *(Member in person)* | **Fully-Dressed** | FR-005, FR-009..FR-011, FR-013 | BR-002, BR-003, BR-006, BR-011, BR-013, BR-014, BR-016, BR-017 | LMS-007 (S2), LMS-017 (S3 hold pickup) |
| UC-06 | **Return a Book Copy** | Librarian | **Fully-Dressed** | FR-005, FR-012..FR-015, FR-022 | BR-006..BR-010, BR-013, BR-017 | LMS-010 (S2), LMS-017 (S3 hold promo) |
| UC-07 | **Renew an Active Loan** | Member, Librarian | Brief | FR-024 | BR-003..BR-006, BR-011 | LMS-014 (S3) |
| UC-08 | **Reserve a Book (Place or Cancel Hold)** | Member, Librarian | Brief | FR-020..FR-023 | BR-011..BR-014 | LMS-017 (S3), LMS-012 (S3) |
| UC-09 | **Pay or Waive Overdue Fine & Assess Lost Copy** | Librarian *(Member)* | **Fully-Dressed** | FR-005, FR-014, FR-015, FR-025..FR-027 | BR-007..BR-011, BR-015, BR-017 | LMS-009 (S2), LMS-011 (S3), LMS-016 (S3) |
| UC-10 | **View Member Self-Service Portal & History** | Member, Librarian | Brief | FR-008, FR-018 | BR-001, BR-002, BR-006, BR-010, BR-013 | LMS-013 (S2), LMS-017 (S3 holds view) |
| UC-11 | **View Operational Dashboard & Export Reports** | Librarian | Brief | FR-014, FR-015, FR-028, FR-029, FR-032 | BR-001, BR-007, BR-008, BR-011 | LMS-015 (S3), LMS-021 (S3), LMS-019 (S3) |

## 1.6 Domain Glossary & Conceptual Domain Model Bridge

Full ubiquitous language glossary (20 entity/policy definitions + 8 enumeration tables): docs/02-inception/DOMAIN_GLOSSARY.md. Conceptual domain model (7 classes, 50 attributes, 6 enums, 8 associations A1..A8, and narrative traceability D1..D8): docs/01-proposal/HW01_Problem_Solution_DomainModel.md.

| Ubiquitous Term | Seed File (data/seed/) | Natural Key & Core Attributes | Key Domain Invariant |
|---|---|---|---|
| **Book** | books-openlibrary.json (782: 779 circulating, 3 referenceOnly) | isbn13 (13-digit ISBN), title, subtitle, authors, publisher, firstPublishYear, category (11), subjects, pageCount, coverId, coverUrl, openLibraryWorkKey, referenceOnly | Owns 1..4 physical BookCopy items (A1). If referenceOnly == true (3 books), copies cannot be loaned or reserved (BR-014). |
| **BookCopy** | copies.json (1,706: 1,624 AVAILABLE, 59 LOANED, 15 MAINTENANCE, 5 LOST, 3 RESERVED) | barcode (LMS-COPY-1001..), isbn13 (FK), status (AVAILABLE, LOANED, RESERVED, MAINTENANCE, LOST) | Governed by BookCopyState; at most one ACTIVE loan (59) or one READY reservation (3) per copy at any moment (BR-017). |
| **Member** | members.json (60: 40 STUDENT, 10 ACADEMIC_STAFF, 10 ADMINISTRATIVE_STAFF; 55 ACTIVE, 5 SUSPENDED) | memberId (SYN-STU-*, SYN-STF-*, SYN-ADM-*), firstName, lastName, email (@synthetic.lms.local), memberType, status (ACTIVE, SUSPENDED), department, registeredOn | memberId represents the personal university ID card (BR-016). Quotas resolved via LoanPolicyStrategy (7 / 3 / 3, BR-002). |
| **Librarian** | librarians.json (3 ROLE_LIBRARIAN) | username (syn-lib-0001..), displayName, email, role (ROLE_LIBRARIAN) | Zero password fields in JSON; BCrypt demo hashes seeded at startup (BR-018). Settles fines (A5). |
| **Loan** | loans.json (164: 59 ACTIVE, 100 RETURNED, 5 LOST) | loanId (LOAN-0001..), memberId, copyBarcode, checkoutDate, dueDate, returnDate, status (ACTIVE, RETURNED, LOST), renewalCount (0..1), renewedOn, lostReportedOn | dueDate = checkoutDate + 30 * (1 + renewalCount). Renewed return triggers 30-day cooldown (BR-006). Lost petition freezes fine (BR-015). |

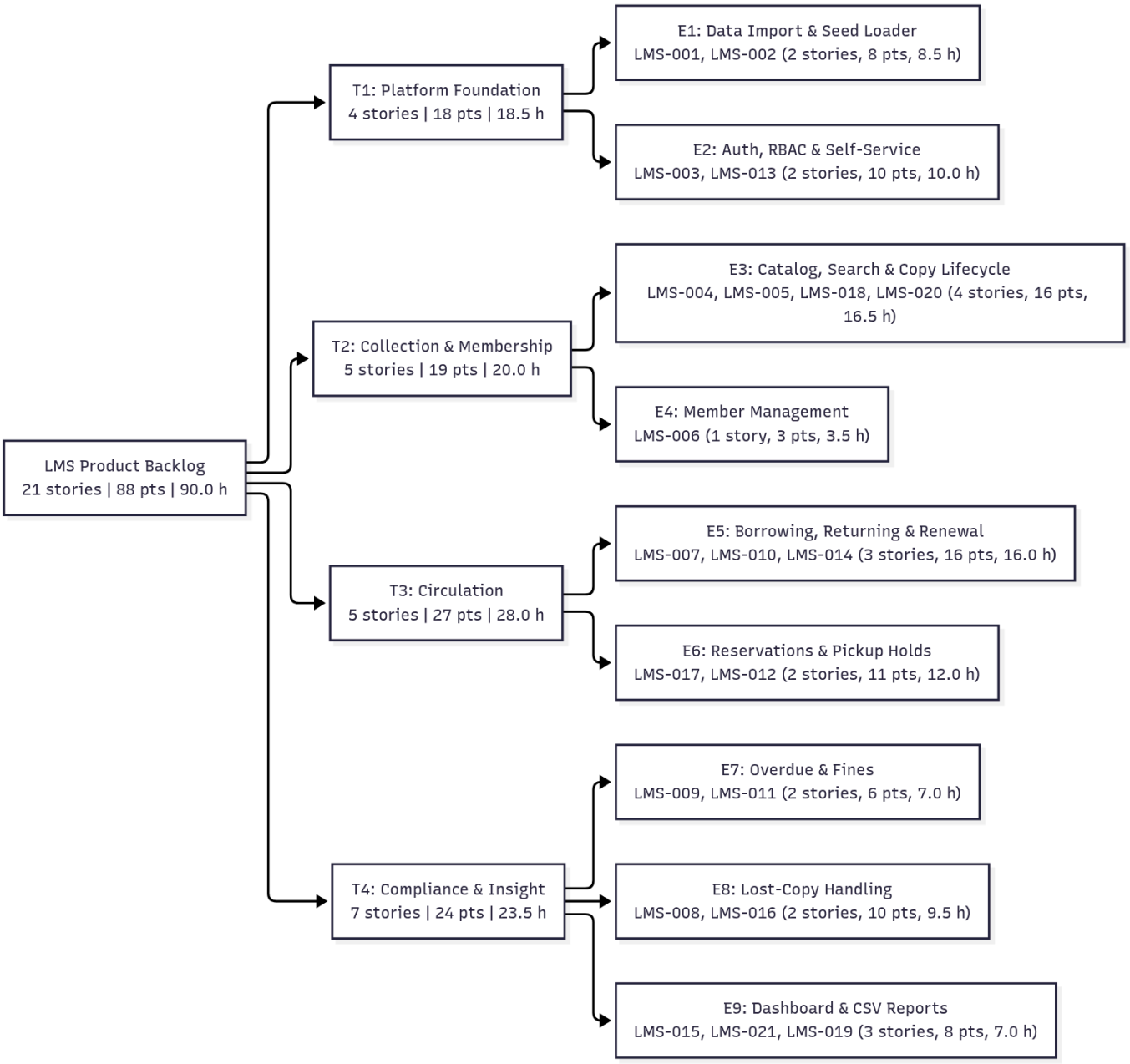
| Ubiquitous Term | Seed File ( data/seed/ ) | Natural Key & Core Attributes | Key Domain Invariant |
|---|---|---|---|
| **Fine** | fines.json ( 47 : 26 UNPAID [ 187.50 TRY ], 16 PAID , 5 WAIVED ) | fineId ( FINE-0001 ..), loanId (unique FK), memberId , overdueDays , amount ( 0.25 * overdueDays ), currency ( TRY ), status ( UNPAID , PAID , WAIVED ), assessedOn , settledOn , settledBy | Uncapped 0.25 TRY/day ( BR-008 , BR-009 ). Any UNPAID fine blocks new checkouts and renewals ( BR-011 ). |
| **Reservation** | reservations.json ( 22 : 19 WAITING , 3 READY ) | reservationId ( RES-0001 ..), isbn13 , memberId , queuePosition ( 1..n ), status ( WAITING , READY ), createdOn , readyCopyBarcode , readyOn , holdUntil | Title-level FIFO queue when 0 copies are AVAILABLE ( BR-012 ); READY hold binds RESERVED copy for 3 days ( holdUntil = readyOn + 3 days , BR-013 ). |

# Part 2 — Agile Scrum Product Backlog & Iteration Plan

## 2.1 Themes & Epics ( T1 – T4 , E1 – E9 )

Full specification: docs/02-inception/THEMES_EPICS.md . The product is organized into **4 Themes** and **9 Epics** decomposing into **21 User Stories** ( 88 Story Points, 173.0 classic hours, 90.0 human hours) and **105 Technical Tasks**.

| Theme | Epic | Goal | Stories | Points | Classic h | Human h | Assigned Story IDs ( Sprint ) |
|---|---|---|---|---|---|---|---|
| **T1 Platform Foundation** | — | Data and access foundations every feature relies on | **4** | **18** | **36** | **18.5** | — |
| ↳ T1 | **E1 Data import & seed loader** | Load seed v2 reproducibly with dates shifted to startup day ( OPEN-04 ). | 2 | 8 | 16 | 8.5 | LMS-001 (S1), LMS-002 (S1) |
| ↳ T1 | **E2 Auth, RBAC & self-service** | Local login with BCrypt demo passwords ( OPEN-05 ), RBAC, member self-service. | 2 | 10 | 20 | 10.0 | LMS-003 (S1), LMS-013 (S2) |
| **T2 Collection & Membership** | — | Who and what the library holds | **5** | **19** | **38** | **20.0** | — |
| ↳ T2 | **E3 Catalog, search & copy lifecycle** | Find books, maintain books/copies, enforce BookCopyState & referenceOnly . | 4 | 16 | 32 | 16.5 | LMS-004 (S1), LMS-005 (S1), LMS-018 (S3), LMS-020 (S3) |
| ↳ T2 | **E4 Member management** | Register, edit, and suspend synthetic members across 3 member types. | 1 | 3 | 6 | 3.5 | LMS-006 (S2) |
| **T3 Circulation** | — | Lending, returning, renewing, and holding books | **5** | **27** | **55** | **28.0** | — |
| ↳ T3 | **E5 Borrowing, returning & renewal** | Desk checkout, return, and single renewal under confirmed ADU rules. | 3 | 16 | 32 | 16.0 | LMS-007 (S2), LMS-010 (S2), LMS-014 (S3) |
| ↳ T3 | **E6 Reservations & pickup holds** | Title-level FIFO holds, in-app notification, 3-day pickup window, cancellation/expiry. | 2 | 11 | 23 | 12.0 | LMS-017 (S3), LMS-012 (S3) |
| **T4 Compliance & Insight** | — | Fines, lost copies, and management information | **7** | **24** | **44** | **23.5** | — |
| ↳ T4 | **E7 Overdue & fines** | Deterministic overdue days, uncapped 0.25 TRY/day fines, desk payment & waiver. | 2 | 6 | 13 | 7.0 | LMS-009 (S2), LMS-011 (S3) |
| ↳ T4 | **E8 Lost-copy handling** | Design spike ( LMS-008 ) + lost-copy petition, 30-day window, 2x runtime appraisedValue . | 2 | 10 | 16 | 9.5 | LMS-008 (S1), LMS-016 (S3) |
| ↳ T4 | **E9 Dashboard & CSV reports** | Operational KPI cards ( 015a ), overdue/unpaid report ( 015b ), CSV export. | 3 | 8 | 15 | 7.0 | LMS-015 (S3), LMS-021 (S3), LMS-019 (S3) |
| **Total** | **9 Epics** | **Complete Product Backlog** | **21** | **88** | **173** | **90.0** | **11 Must, 7 Should, 3 Could** |




## 2.2 User Stories, Estimation Model & Task Breakdown ( `LMS-001` .. `LMS-021` )

Full Given/When/Then acceptance criteria for all 21 stories: `docs/02-inception/USER_STORIES.md` . Full breakdown of all 105 technical tasks: `docs/02-inception/TASKS.md` .

### Estimation Methodology (Product Owner Approved)

1. **Fibonacci Story Points ( `1, 2, 3, 5, 8` ; total `88 pts` ):** Relative complexity, rule density, and cross-entity touchpoints ( `Must = 48 pts` , `Should = 32 pts` , `Could = 8 pts` ). Estimates re-calibrated in commit `df7eaa4` from the initial 20-story draft ( `6b4265c` / `28c40d9` ), see reconciliation note in `docs/process/BACKLOG_POINT_RECONCILIATION.md` ; the dashboard story was split into `LMS-015` ( `015a` KPI cards, `2 pts` ) and `LMS-021` ( `015b` overdue/unpaid report table, `3 pts` ) so the commitment line lands cleanly inside Sprint 3 net capacity.
2. **Classic Effort Hours ( `Classic h` ; total `173.0 h` , Must `98.0 h` ):** Traditional two-student implementation time without AI coding assistants ( `~2.0 h/pt` ).
3. **Human Effort Hours ( `Human h` ; total `90.0 h` , Must `53.0 h` , Should `29.5 h` , Could `7.5 h` ):** The actual human time of the two students—writing prompts, steering AI agents, reviewing generated code/docs, executing manual and automated tests, and reviewing Stitch UI screens. Agent runtime is **not** counted. Calculated from `~0.40 x` classic coding effort plus fixed human review/testing overhead, a `4.0 h` timeboxed design spike ( `LMS-008` ), and PO-approved safety reserves on the two most rule-dense stories ( `LMS-005 = 7.5 h` , `LMS-007 = 9.0 h` ). **All estimates are explicit planning assumptions to be re-calibrated with Sprint 1 actual hours.**
4. **Agent Sessions ( `46` sessions):** Budgeted heavy AI coding/design sessions ( `1..4` per story) tracked against weekly AI quota limits ( `RISK-H03` ).
5. **Task Breakdown ( `105` tasks = `90.0` human h in `docs/02-inception/TASKS.md` ):** Every story is decomposed into `1.0 – 4.0 h` technical tasks covering Stitch MCP UI prototyping ( `NFR-001` , scheduled strictly before Thymeleaf view tasks),

domain/service/repository coding, and JUnit 5 TDD unit/integration tests ( NFR-011 ), with an empty Actual human h column ready for Sprint 1–3 tracking.

### Catalog of User Stories ( LMS-001 .. LMS-021 )

| ID | Story Title & User Story Statement (*As a … I want … so that …*) | Key Acceptance Criteria Highlights ( Given / When / Then ) | Traceability ( FR / UC / BR ) | Est. ( Pts / Classic / Human h / Tasks ) |
|---|---|---|---|---|
| LMS-001 | **Domain model and persistence skeleton:** As a developer, I want the 7 JPA entities ( Book , BookCopy , Member , Librarian , Loan , Fine , Reservation ), 6 enums, and Spring Data repositories mapped to H2 so that all stories build on a tested persistence layer. | Schema starts cleanly on file H2 ( ./data/db/lms ) and in-memory test H2 ( jdbc:h2:mem:lms_test ); Book ( 9780002615075 ) with BookCopy ( LMS-COPY-1002 ), Member ( SYN-STU-0001 ), and Loan ( LOAN-0099 ) round-trip with all associations ( A1 .. A8 ); unique barcode and Fine.loanId constraints reject duplicates. | FR-001 , FR-004 .. FR-006 , FR-011 , FR-015 , FR-019 · BR-010 , BR-017 | 3 pts / 6 h / 3.5 h ( 5 tasks) |
| LMS-002 | **Seed loader with date shifting ( OPEN-04 ):** As a librarian/evaluator, I want the app to load data/seed/*.json on startup when H2 is empty and shift all seed dates by delta = today - 2026-10-22 so that active, due-soon, and overdue scenarios stay fresh on any demo date. | Empty H2 loads exact seed v2 counts ( 782 books with 3 referenceOnly , 1,706 copies [ 1,624 AVAILABLE , 59 LOANED , 15 MAINTENANCE , 5 LOST , 3 RESERVED ], 60 members [ 40/10/10 ; 55 ACTIVE , 5 SUSPENDED ], 3 librarians, 164 loans [ 59 ACTIVE , 100 RETURNED , 5 LOST ], 47 fines [ 26 UNPAID totaling 187.50 TRY , 16 PAID , 5 WAIVED ], 22 reservations [ 19 WAITING , 3 READY ]); non-empty H2 skips reload (idempotent); date shift preserves overdueDays ( FINE-0022 on LOAN-0107 ) and holdUntil - readyOn == 3 days ( RES-0009 ). | FR-019 , DR-001 .. DR-007 · BR-017 , BR-018 | 5 pts / 10 h / 5.0 h ( 5 tasks) |
| LMS-003 | **Login/logout with BCrypt demo passwords ( OPEN-05 ) and RBAC:** As a librarian or member, I want to log in with a startup-seeded BCrypt password and see only endpoints permitted for my role so that desk operations are protected. | data/seed/*.json contains zero password fields; startup seeds BCrypt hashes for syn-lib-0001 .. 0003 ( ROLE_LIBRARIAN ) and SYN-STU-* / SYN-STF-* / SYN-ADM-* ( ROLE_MEMBER ); ROLE_MEMBER gets HTTP 403 on librarian CRUD/desk URLs; invalid credentials show safe error; logout invalidates session + CSRF. | FR-016 , FR-017 , NFR-007 , NFR-008 · UC-02 · BR-001 , BR-018 | 5 pts / 10 h / 5.0 h ( 6 tasks) |
| LMS-004 | **Catalog search and book detail:** As a member or librarian, I want to search books by partial title, author, or exact isbn13 , filter by category, availability, or referenceOnly , and inspect copy states on the detail page. | Query "Yaşar Kemal" or "9780002615075" returns *İnce Memed* with available/total copy counts; referenceOnly == true ( 9780824722876 ) shows REFERENCE ONLY — NOT FOR LOAN badge and disables borrow/reserve; detail page lists all copies ( LMS-COPY-1002 .. 1004 ) and active hold queue count. | FR-002 , FR-003 · UC-01 · BR-001 , BR-014 , BR-017 | 3 pts / 6 h / 3.5 h ( 5 tasks) |
| LMS-005 | **Book CRUD, copy registration and copy lifecycle:** As a librarian, I want to create/edit/archive Book records, register LMS-COPY-XXXX barcodes, and enforce BookCopyState transitions so that inventory stays accurate. | Invalid ISBN-13 checksum is rejected; valid book creation persists all fields including referenceOnly ; new copy accepts AVAILABLE / MAINTENANCE and rejects duplicate barcode; BookCopyState allows AVAILABLE <-> MAINTENANCE and blocks LOANED -> MAINTENANCE or checkout of LOANED / MAINTENANCE / LOST . | FR-001 , FR-004 , FR-005 · UC-03 · BR-014 , BR-015 , BR-017 | 8 pts / 16 h / 7.5 h ( 7 tasks) |
| LMS-008 | **Design spike — compensation tracking & reservation terminal states:** As an architect, I want a timeboxed spike resolving HW01 §3.7 open design items before Sprint 3 coding starts so that LMS-016 and LMS-017 require zero schema rework. | Produces ADR in docs/03-design/ specifying (a) how 2.00 * appraisedValue lost-copy compensation is persisted and blocks borrowing under BR-011 , and (b) whether fulfilled/cancelled/expired reservations use terminal enum values or deletion; updates JPA skeleton with green tests. | FR-011 , FR-023 , FR-027 · UC-05 , UC-08 , UC-09 · BR-011 .. BR-013 , BR-015 | 2 pts / 4 h / 4.0 h ( 3 tasks) |
| LMS-006 | **Member registration, profile update and suspension:** As a librarian, I want to register synthetic STUDENT , ACADEMIC_STAFF , and ADMINISTRATIVE_STAFF members, edit profiles, and toggle ACTIVE / SUSPENDED status. | Validates SYN-STU-* / SYN-STF-* / SYN-ADM-* ID and @synthetic.lms.local email ( NFR-009 ); displays confirmed quota ( 7 / 3 / 3 ); toggling status to SUSPENDED immediately blocks new checkouts, renewals, and reservations ( BR-011 ). | FR-006 , FR-007 · UC-04 · BR-001 , BR-002 , BR-011 , BR-018 | 3 pts / 6 h / 3.5 h ( 5 tasks) |
| LMS-007 | **Checkout with eligibility, cooldown and referenceOnly enforcement:** As a librarian, I want to check out an eligible copy to an in-person member ( memberId + copyBarcode ) with full validation of ADU rules in one @Transactional step. | LoanPolicyStrategy enforces max 7 ( ACADEMIC_STAFF ) vs 3 ( ADMINISTRATIVE_STAFF / STUDENT ) and dueDate = checkoutDate + 30 days ; blocks SUSPENDED , overdue active loan ( LOAN-0099 ), UNPAID fine ( FINE-0022 ), referenceOnly == true ( 9780824722876 ), non-AVAILABLE copy, or active 30-day post-renewal cooldown ( BR-006 ). | FR-009 .. FR-011 , FR-013 · UC-05 · BR-002 , BR-003 , BR-006 , BR-011 , BR-013 , BR-014 , BR-016 , BR-017 | 8 pts / 18 h / 9.0 h ( 7 tasks) |

| ID | Story Title & User Story Statement (*As a … I want … so that …*) | Key Acceptance Criteria Highlights (Given / When / Then) | Traceability (FR / UC / BR) | Est. (Pts / Classic / Human h / Tasks) |
|---|---|---|---|---|
| LMS-009 | **Overdue detection and uncapped fine calculation:** As a librarian, I want deterministic overdue days (effectiveEndDate > dueDate, zero grace) and uncapped 0.25 TRY/day fines (FineCalculationStrategy) so penalties match ADU rules. | effectiveEndDate == dueDate yields 0 overdue days and no fine; 1 day late yields 0.25 TRY UNPAID; 1,200 days late yields 300.00 TRY (no 250 TRY cap, BR-009); lostReportedOn freezes overdueDays as of petition date (LOAN-0103, BR-015). | FR-014, FR-015 · UC-06, UC-09, UC-11 · BR-007..BR-010, BR-015 | 3 pts / 6 h / 3.5 h (4 tasks) |
| LMS-010 | **Return of a copy (AVAILABLE-only in Sprint 2):** As a librarian, I want to scan a returned copyBarcode to close its ACTIVE loan, finalize any overdue fine, record post-renewal cooldown anchor, and transition the copy to AVAILABLE. | Atomic @Transactional return sets returnDate = currentDate, status = RETURNED, transitions copy -> AVAILABLE (hold promotion to RESERVED is added in Sprint 3 by LMS-017), persists/updates UNPAID fine if returnDate > dueDate, and anchors 30-day cooldown if renewalCount >= 1. | FR-012..FR-015 · UC-06 · BR-006..BR-010, BR-017 | 3 pts / 6 h / 3.5 h (5 tasks) |
| LMS-013 | **Member self-service portal & librarian member inspection:** As a member (or librarian inspecting a member), I want to view active loans, due dates, historical loans, 30-day cooldown timers, and fines (UNPAID / PAID / WAIVED). | Authenticated member (SYN-STU-0001) sees own active/historical loans, cooldown end dates, and fines (LOAN-0099) and cannot access another member's profile (403); Librarian (syn-lib-0001) inspects any member with quota (7 / 3 / 3) and block reason banner. *(Holds/notifications added in LMS-017).* | FR-008, FR-018 · UC-04, UC-10 · BR-001, BR-002, BR-006, BR-010, BR-011 | 5 pts / 10 h / 5.0 h (6 tasks) |
| LMS-017 | **Reservation, FIFO queue, hold promotion & in-app notification:** As a member or librarian, I want to place a FIFO hold when 0 copies are AVAILABLE, and have a returned copy automatically held as RESERVED (READY) for 3 days with an in-app alert. | Rejects hold if >= 1 copy is AVAILABLE, referenceOnly == true, member blocked (BR-011), or already borrowing/queuing isbn13; assigns FIFO queuePosition = 1..n; on return (UC-06 Observer), promotes queuePosition == 1 to READY (readyCopyBarcode, holdUntil = readyOn + 3 days), sets copy -> RESERVED, creates in-app notification, and allows checkout of RESERVED copy only to queuePosition == 1 member. | FR-010..FR-012, FR-018, FR-020..FR-022 · UC-05, UC-06, UC-08, UC-10 · BR-011..BR-014, BR-017 | 8 pts / 16 h / 8.5 h (7 tasks) |
| LMS-011 | **Fine payment (PAID) and waiver (WAIVED):** As a librarian, I want to record desk payment or administrative waiver of an UNPAID fine on a resolved loan so that the member's financial block is cleared with auditability. | Marks Fine.status = PAID or WAIVED with settledOn = currentDate and settledBy = librarian.username (syn-lib-0001); rejects settlement while loan is still ACTIVE; clears BR-011 block once all overdue loans and UNPAID balances reach zero. | FR-025, FR-026 · UC-09 · BR-010, BR-011 | 3 pts / 7 h / 3.5 h (5 tasks) |
| LMS-012 | **Reservation cancellation and 3-day hold expiry:** As a member or librarian, I want to cancel a reservation and have unclaimed READY holds expire after 3 days (currentDate > holdUntil) so RESERVED copies are never locked forever. | Cancelling WAITING decrements trailing queuePositions; cancelling or expiring a READY hold promotes the next WAITING reservation (rebinding readyCopyBarcode and holdUntil = currentDate + 3 days) or returns copy.status to AVAILABLE if queue is empty. | FR-023 · UC-08 · BR-012, BR-013, BR-017 | 3 pts / 7 h / 3.5 h (5 tasks) |
| LMS-015 | **Operational dashboard KPI summary cards (015a):** As a librarian, I want a dashboard with real-time KPI cards (titles, copies by state, active loans, due-soon 0..3 days, overdue loans, total UNPAID fine TRY) for instant health visibility. | Seeded dataset renders 782 titles (3 referenceOnly), 1,706 copies across all 5 states (1,624 AVAILABLE, 59 LOANED, 15 MAINTENANCE, 5 LOST, 3 RESERVED), 59 active loans, 12 due-soon (dueDate in [currentDate, currentDate + 3 days]), 15 overdue active loans, and 187.50 TRY across 26 UNPAID fines. | FR-028 · UC-11 · BR-001, BR-007, BR-008 | 2 pts / 4 h / 2.0 h (4 tasks) |
| LMS-014 | **Loan renewal with reserved guard (Stretch):** As a member or librarian, I want to renew an eligible ACTIVE loan once (+30 days) when not overdue, not already renewed, and not reserved by another member. | renewalCount: 0 -> 1, renewedOn = currentDate, dueDate = checkoutDate + 60 days; rejects if renewalCount >= 1, currentDate > dueDate, any WAITING / READY reservation exists on isbn13, or borrower is blocked (BR-011). | FR-024 · UC-07 · BR-003..BR-006, BR-011 | 5 pts / 8 h / 3.5 h (5 tasks) |
| LMS-016 | **Lost-copy process with runtime appraisedValue (Stretch):** As a librarian, I want to record a lost-book petition (lostReportedOn), freeze daily overdue fines, and after 30 days enter appraisedValue to bill 2.00 * appraisedValue. | Recording lostReportedOn freezes overdueDays; replacement within 30 days closes loan as RETURNED; after 30 days, Librarian enters appraisedValue (e.g., 450.00 TRY), system charges 900.00 TRY (2x) + frozen fine, sets copy -> LOST and loan -> LOST (returnDate = null). | FR-005, FR-014, FR-027 · UC-09 · BR-007, BR-011, BR-015, BR-017 | 8 pts / 12 h / 5.5 h (6 tasks) |

| ID | Story Title & User Story Statement (*As a … I want … so that …*) | Key Acceptance Criteria Highlights (`Given / When / Then`) | Traceability (`FR / UC / BR`) | Est. (`Pts / Classic / Human h / Tasks`) |
|---|---|---|---|---|
| `LMS-021` | **Actionable overdue loans & unpaid fines report (`015b`, *Stretch*):** As a librarian, I want a prioritized table of overdue active loans and `UNPAID` fines with direct links to return/settle so I can act on delinquencies. | Displays `loanId`, `memberId`, member name, `copyBarcode`, `title`, `dueDate`, `overdueDays` (descending), accrued `amount` (`0.25 TRY/day`), and quick action links to member profile, return desk, and fine settlement. | `FR-014`, `FR-015`, `FR-029` · `UC-11` · `BR-001`, `BR-007`, `BR-008`, `BR-011` | `3 pts` / `6 h` / `3.0 h` (`3` tasks) |
| `LMS-018` | **Public guest catalog portal (`C1`, *Stretch*):** As an unauthenticated guest (`ROLE_GUEST`), I want to browse/search the catalog and view copy availability in read-only mode without seeing any member or loan data. | Anonymous `GET /catalog` and `/catalog/{isbn13}` succeed (`200 OK`) showing availability and `referenceOnly` badges; zero member/loan PII is rendered; reserve/borrow buttons redirect to `/login`. | `FR-002`, `FR-003`, `FR-030` · `UC-01` · `BR-001`, `BR-014` | `2 pts` / `4 h` / `2.0 h` (`4` tasks) |
| `LMS-019` | **CSV export of overdue & inventory reports (`C3`, *Stretch*):** As a librarian, I want to download the overdue loans report and physical copy inventory as UTF-8 CSV files for offline auditing. | `ROLE_LIBRARIAN` downloads valid UTF-8 CSV (`text/csv`) with escaped fields and Turkish diacritics (`ğ, ş, ı, ö, ü, ç`) intact; non-librarians receive HTTP `403`. | `FR-032` · `UC-11` · `BR-001` | `3 pts` / `5 h` / `2.0 h` (`4` tasks) |
| `LMS-020` | **Open Library ISBN lookup helper (`C2`, *Stretch*):** As a librarian adding a book, I want an optional "Fetch by ISBN" helper (`OpenLibraryCatalogAdapter`) that pre-fills metadata and degrades gracefully when offline. | Online lookup populates title, authors, publisher, year, and cover URL; offline or timeout (`<= 3 s`) shows a non-blocking banner (*"Offline mode: please enter book metadata manually"*) keeping manual form entry 100% functional (`NFR-004`). | `FR-001`, `FR-031` · `UC-03` · `BR-001` | `3 pts` / `6 h` / `3.5 h` (`4` tasks) |

## 2.3 Ordered Product Backlog & End-to-End Traceability

Authoritative source: `docs/02-inception/PRODUCT_BACKLOG.md`, validated by `python3 tools/agile/validate_backlog.py`.

### Ordered Product Backlog (`LMS-001` .. `LMS-021`)

| Order | ID | Epic | Story Title | MoSCoW | Pts | Classic h | Human h | Agent Sess. | Sprint | Depends On | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | `LMS-001` | E1 | Domain model and persistence skeleton | Must | 3 | 6 | 3.5 | 2 | S1 | — | Planned (S1) |
| 2 | `LMS-002` | E1 | Seed loader with date shifting | Must | 5 | 10 | 5.0 | 3 | S1 | `LMS-001` | Planned (S1) |
| 3 | `LMS-003` | E2 | Login/logout with BCrypt demo passwords and RBAC | Must | 5 | 10 | 5.0 | 3 | S1 | `LMS-001`, `LMS-002` | Planned (S1) |
| 4 | `LMS-004` | E3 | Catalog search and book detail | Must | 3 | 6 | 3.5 | 2 | S1 | `LMS-002`, `LMS-003` | Planned (S1) |
| 5 | `LMS-005` | E3 | Book CRUD, copy registration and copy lifecycle | Must | 8 | 16 | 7.5 | 3 | S1 | `LMS-003`, `LMS-004` | Planned (S1) |
| 6 | `LMS-008` | E8 | Design spike: compensation tracking and reservation terminal states | Must | 2 | 4 | 4.0 | 1 | S1 | `LMS-001` | Planned (S1) |
| 7 | `LMS-006` | E4 | Member registration, profile update and suspension | Must | 3 | 6 | 3.5 | 2 | S2 | `LMS-003` | Planned (S2) |
| 8 | `LMS-007` | E5 | Checkout with eligibility, cooldown and referenceOnly enforcement | Must | 8 | 18 | 9.0 | 4 | S2 | `LMS-005`, `LMS-006` | Planned (S2) |
| 9 | `LMS-009` | E7 | Overdue detection and uncapped fine calculation | Must | 3 | 6 | 3.5 | 2 | S2 | `LMS-001` | Planned (S2) |
| 10 | `LMS-010` | E5 | Return of a copy (copy to AVAILABLE only) | Must | 3 | 6 | 3.5 | 2 | S2 | `LMS-007`, `LMS-009` | Planned (S2) |
| 11 | `LMS-013` | E2 | Member self-service portal and librarian member inspection | Must | 5 | 10 | 5.0 | 3 | S2 | `LMS-007`, `LMS-009`, `LMS-010` | Planned (S2) |
| 12 | `LMS-017` | E6 | Reservation, FIFO queue, hold promotion and in-app notification | Should | 8 | 16 | 8.5 | 4 | S3 | `LMS-010`, `LMS-008` | Planned (S3: **commit**) |

| Order | ID | Epic | Story Title | MoSCoW | Pts | Classic h | Human h | Agent Sess. | Sprint | Depends On | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 13 | LMS-011 | E7 | Fine payment and waiver | Should | 3 | 7 | 3.5 | 2 | S3 | LMS-009, LMS-010 | Planned (S3: **commit**) |
| 14 | LMS-012 | E6 | Reservation cancellation and hold expiry | Should | 3 | 7 | 3.5 | 2 | S3 | LMS-017 | Planned (S3: **commit**) |
| 15 | LMS-015 | E9 | Operational dashboard KPI summary cards (015a) | Should | 2 | 4 | 2.0 | 1 | S3 | LMS-009, LMS-010 | Planned (S3: **commit**) |
| 16 | LMS-014 | E5 | Loan renewal with reserved guard | Should | 5 | 8 | 3.5 | 2 | S3 | LMS-007 | Planned (S3: *stretch*) |
| 17 | LMS-016 | E8 | Lost-copy process with runtime appraised value | Should | 8 | 12 | 5.5 | 3 | S3 | LMS-010, LMS-011, LMS-008 | Planned (S3: *stretch*) |
| 18 | LMS-021 | E9 | Actionable overdue loans and unpaid fines report (015b) | Should | 3 | 6 | 3.0 | 1 | S3 | LMS-009, LMS-010, LMS-015 | Planned (S3: *stretch*) |
| 19 | LMS-018 | E3 | Public guest catalog portal | Could | 2 | 4 | 2.0 | 1 | S3 | LMS-004 | Planned (S3: *stretch*) |
| 20 | LMS-019 | E9 | CSV export of overdue and inventory reports | Could | 3 | 5 | 2.0 | 1 | S3 | LMS-021 | Planned (S3: *stretch*) |
| 21 | LMS-020 | E3 | Open Library ISBN lookup helper | Could | 3 | 6 | 3.5 | 2 | S3 | LMS-005 | Planned (S3: *stretch*) |
| **Total** | **21** | **E1–E9** | **11 Must (48 pts / 53.0 h), 7 Should (32 pts / 29.5 h), 3 Could (8 pts / 7.5 h)** | — | **88** | **173** | **90.0** | **46** | **S1–S3** | **0 forward deps** | **105 tasks (90.0 h)** |

### End-to-End Traceability Matrix (FR-001..FR-032 -> UC -> BR -> Story -> Sprint)

Every Must Have functional requirement (FR-001..FR-019) is assigned to a story in Sprint 1 or Sprint 2 (Must FR not covered in S1+S2: none).

| FR-* ID | MoSCoW Priority | Realizing Use Case(s) | Enforced Business Rule(s) | Implementing Story | Target Sprint |
|---|---|---|---|---|---|
| FR-001, FR-004, FR-005 | Must Have (M1) | UC-03, UC-05, UC-06, UC-09 | BR-014, BR-015, BR-017 | LMS-005 | **S1** |
| FR-002, FR-003 | Must Have (M1) | UC-01 | BR-001, BR-014, BR-017 | LMS-004 | **S1** |
| FR-006, FR-007 | Must Have (M2) | UC-04 | BR-001, BR-002, BR-011, BR-018 | LMS-006 | **S2** |
| FR-008, FR-018 | Must Have (M2, M5) | UC-04, UC-10 | BR-001, BR-002, BR-006, BR-010, BR-011, BR-013 | LMS-013 | **S2** |
| FR-009, FR-010, FR-011, FR-013 | Must Have (M3) | UC-05, UC-06 | BR-002, BR-003, BR-006, BR-011, BR-013, BR-014, BR-016, BR-017 | LMS-007 | **S2** |
| FR-012 | Must Have (M3) | UC-06 | BR-007, BR-008, BR-013, BR-017 | LMS-010 | **S2** |
| FR-014, FR-015 | Must Have (M4) | UC-06, UC-09, UC-11 | BR-007, BR-008, BR-009, BR-010, BR-015 | LMS-009 | **S2** |
| FR-016, FR-017 | Must Have (M5) | UC-02 | BR-001, BR-018 | LMS-003 | **S1** |
| FR-019 | Must Have (M6) | System Startup | BR-017, BR-018 | LMS-002 (on LMS-001) | **S1** |
| FR-020, FR-021, FR-022 | Should Have (S1) | UC-06, UC-08, UC-10 | BR-011, BR-012, BR-013, BR-014 | LMS-017 | **S3 (Commit)** |
| FR-023 | Should Have (S1) | UC-08 | BR-012, BR-013 | LMS-012 | **S3 (Commit)** |

| FR-* ID | MoSCoW Priority | Realizing Use Case(s) | Enforced Business Rule(s) | Implementing Story | Target Sprint |
|---|---|---|---|---|---|
| FR-024 | Should Have (S2) | UC-07 | BR-003, BR-004, BR-005, BR-011 | LMS-014 | **S3 (Stretch)** |
| FR-025, FR-026 | Should Have (S3) | UC-09 | BR-010, BR-011 | LMS-011 | **S3 (Commit)** |
| FR-027 | Should / Could (S3) | UC-09 | BR-005, BR-015, BR-017 | LMS-016 (on LMS-008) | **S3 (Stretch)** |
| FR-028 | Should Have (S4) | UC-11 | BR-001, BR-007, BR-008 | LMS-015 (015a) | **S3 (Commit)** |
| FR-029 | Should Have (S4) | UC-11 | BR-007, BR-008, BR-011 | LMS-021 (015b) | **S3 (Stretch)** |
| FR-030 | Could Have (C1) | UC-01 | BR-001, BR-014 | LMS-018 | **S3 (Stretch)** |
| FR-031 | Could Have (C2) | UC-03 | BR-001 | LMS-020 | **S3 (Stretch)** |
| FR-032 | Could Have (C3) | UC-11 | BR-001 | LMS-019 | **S3 (Stretch)** |

## 2.4 Three-Sprint Iteration Plan, Capacity Model & Commitment Line

Authoritative source: `docs/02-inception/SPRINT_PLAN.md`.

### Sprint Schedule & Capacity Model

- **Capacity Unit (`Human h`):** Two students' human effort (`8 h/person/week` in normal weeks, reduced to `3 h/person/week` during midterm weeks `Nov 17 – Nov 28`, prorated per calendar day).
- **Fixed Deductions per Sprint:** Scrum ceremonies = `4.0 h` (`2 people x 2 h`); Homework engineering/design documentation = `8.0 h` in Sprint 1 (`HW03`: Use Case diagram `1.5 h`, System Sequence Diagrams `2.0 h`, Class Diagram `2.0 h`, TDD/Git/Maven plan `1.0 h`, Stitch UI review `1.5 h`), `4.0 h` in Sprint 2 (`HW04`), and `7.0 h` in Sprint 3 (`HW05`: user guide, test report, final retrospective); Contingency buffer = **`10%` of gross capacity** (never reduced).
- **Sign Convention for `Planned - Net`:** Positive (`+`) = overrun / over net capacity (`over`); Negative (`-`) = unused net capacity (`slack`).

| Sprint | Calendar Window | Course Deliverable | Gross (h) | Ceremonies (h) | HW Docs (h) | 10% Buffer (h) | Net Story Capacity (h) | Planned Stories (h) | Must (h) | Should (h) | Could (h) | Planned - Net |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Sprint 1** | Nov 11 – Dec 08 (28 d, incl. midterms Nov 17–28) | `HW03` (Elaboration + S1) | `46.9` | `4.0` | `8.0` | `4.7` | **`30.2`** | `28.5` (`26 pts`) | `28.5` | `0.0` | `0.0` | `-1.7 h (slack)` |
| **Sprint 2** | Dec 09 – Dec 22 (14 d) | `HW04` (Construction + S2) | `32.0` | `4.0` | `4.0` | `3.2` | **`20.8`** | `24.5` (`22 pts`) | `24.5` | `0.0` | `0.0` | `+3.7 h (over)` |
| **Sprint 3** | Dec 23 – Jan 05 (14 d) | `HW05` (Transition + MVP) | `32.0` | `4.0` | `7.0` | `3.2` | **`17.8`** | `17.5` committed (`+19.5` stretch) | `0.0` | `29.5` | `7.5` | `-0.3 h (slack, committed) / +19.2 h (over w/ stretch)` |
| **Total** | **Nov 11 – Jan 05** | **`HW03 – HW05`** | **`110.9`** | **`12.0`** | **`19.0`** | **`11.1`** | **`68.8`** | **`70.5` committed (`90.0` all)** | **`53.0`** | **`29.5`** | **`7.5`** | **S1+S2 Must: `+2.0 h` (over)** |

### Must Gap Analysis (`+2.0 h`, `4.0%` of S1+S2 Net Capacity — Accepted Risk `RISK-H01`)

All `11` Must stories require **`53.0` human hours** (`28.5 h` in S1 + `24.5 h` in S2) against a combined S1+S2 net story capacity of **`51.0` hours** (`30.2 + 20.8 h`), leaving a **known gap of `+2.0 h` (`4.0%` of net)**. Rather than artificially lowering the estimates of complex stories (`LMS-005 7.5 h`, `LMS-007 9.0 h`) or cutting the 10% buffer, the Product Owner approved three governance rules:

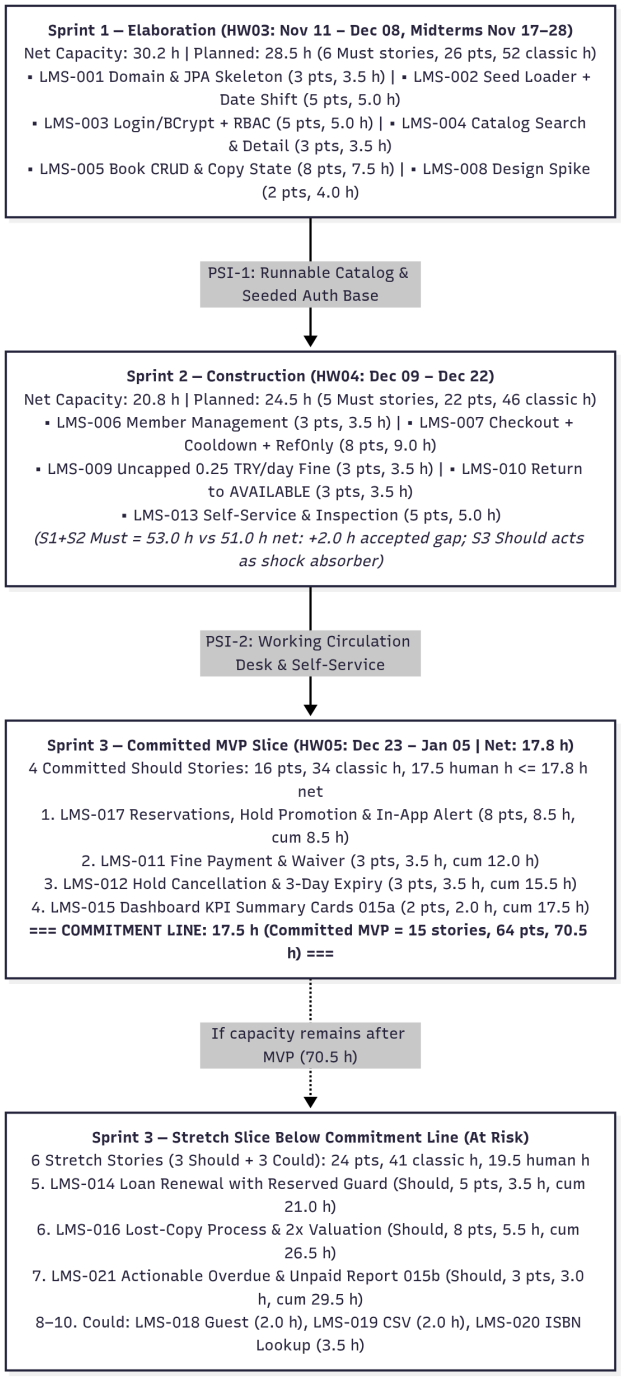
1. **Buffer Preservation:** The 10% buffer (`4.7 h` in S1, `3.2 h` in S2, `3.2 h` in S3) is **not** reduced.
2. **Sprint 3 Should Stories as Shock Absorber:** If Must stories overrun in Sprint 2, the overrun is absorbed by deferring Sprint 3 Should stories from the bottom of the commitment line upward.

3. **Quantitative Re-Plan Trigger:** If at the end of Sprint 1 the actual ratio `actual_human_h / classic_h` exceeds **`0.45`**, or if the remaining Sprint 2 Must estimate exceeds Sprint 2 net capacity (`20.8 h`) by more than **`10%`**, the Sprint 2 plan is re-baselined and any scope decision is brought to the Product Owner.

### Sprint 3 Value Ordering & Commitment Line (**`17.5 h <= 17.8 h`** Net Capacity)

Because total Sprint 3 Should (`29.5 h`) + Could (`7.5 h`) = `37.0 h` exceeds Sprint 3 net capacity (`17.8 h`), the Product Owner ordered the Sprint 3 stories by domain value and state-machine consistency and drew an explicit **commitment line at `17.5 h`**:

| S3 Rank | Story ID & Title | MoSCoW | Pts | Classic h | Human h | Cumulative Human h | Commitment Status & Domain Rationale |
|---|---|---|---|---|---|---|---|
| **1** | `LMS-017` Reservation, FIFO queue, hold promotion & in-app notification | Should (`S1`) | 8 | 16 | `8.5` | `8.5 h` | **COMMIT** — Core FIFO hold queue, `RESERVED` copy promotion on return, and 3-day pickup notification (`FR-020`..`FR-022`, `BR-012`, `BR-013`). |
| **2** | `LMS-011` Fine payment and waiver | Should (`S3`) | 3 | 7 | `3.5` | `12.0 h` | **COMMIT** — Desk `PAID`/`WAIVED` settlement (`FR-025`, `FR-026`, `BR-010`). Required so members with fines accrued in S2 can clear their `BR-011` borrowing block. |
| **3** | `LMS-012` Reservation cancellation and hold expiry | Should (`S1`) | 3 | 7 | `3.5` | `15.5 h` | **COMMIT** — Cancellation & 3-day hold expiry reallocation (`FR-023`). Required for reservation state-machine completeness so an unclaimed `READY` hold never locks a `RESERVED` copy forever. |
| **4** | `LMS-015` Operational dashboard KPI summary cards (`015a`) | Should (`S4`) | 2 | 4 | `2.0` | **`17.5 h <= 17.8 h`** | **COMMIT (Commitment Line)** — Delivers real-time librarian KPI cards (`FR-028`: titles, copies by state, active/due-soon/overdue loans, unpaid fine total). |
| **5** | `LMS-014` Loan renewal with reserved guard | Should (`S2`) | 5 | 8 | `3.5` | `21.0 h` | *STRETCH (below line)* — Single `+30`-day self-service/desk renewal (`FR-024`, `BR-004`, `BR-005`). |
| **6** | `LMS-016` Lost-copy process with runtime `appraisedValue` | Should (`S3`) | 8 | 12 | `5.5` | `26.5 h` | *STRETCH (below line)* — Lost-book petition fine freeze, 30-day window, and `2.00 * appraisedValue` runtime charge (`FR-027`, `BR-015`). |
| **7** | `LMS-021` Actionable overdue loans & unpaid fines report (`015b`) | Should (`S4`) | 3 | 6 | `3.0` | `29.5 h` | *STRETCH (below line)* — Detailed overdue/unpaid report table with quick desk links (`FR-029`). |
| **8–10** | `LMS-018` Guest portal (`2.0 h`), `LMS-019` CSV export (`2.0 h`), `LMS-020` ISBN lookup (`3.5 h`) | Could (`C1`–`C3`) | 8 | 15 | `7.5` | `37.0 h` | *STRETCH (below line)* — Optional enhancements (`FR-030`..`FR-032`) attempted only if all committed and Should stretch items finish early. |




## Cross-Sprint Dependency Fixes, DoR, DoD & Risk Register

- **Dependency Hygiene (Verified by `validate_backlog.py` -> `OVERALL: PASS`):** No story depends on a story in a later sprint. Specifically: (1) `LMS-010` (S2 Return) transitions returned copies to `AVAILABLE` only, while `RESERVED` hold promotion is added in S3 by `LMS-017`; (2) `LMS-013` (S2 Self-Service) displays active/historical loans, cooldowns, and fines only, while holds and in-app notifications are added in S3 by `LMS-017`; (3) `LMS-014` (S3 Renewal) checks active reservations via a repository query; and (4) `LMS-008` (S1 Spike) resolves compensation and reservation terminal-state design before S3.
- **Definition of Ready (DoR):** Standard user story format + testable Given/When/Then criteria with `FR` / `UC` / `BR` tags; Fibonacci points, classic/human hours, and agent sessions estimated; zero forward dependencies; Stitch MCP UI task scheduled before any Thymeleaf task; required seed test data identified in `data/seed/*.json`.
- **Definition of Done (DoD):** `./mvnw -q test` (or `tools/mvnw-vm.sh -q test` on the VM) and `python3 -m unittest discover -s tools/data/tests` pass 100% offline; mandatory CSE305 header on all first-party files; `docs/process/ai-usage-log.md` updated with exact model selector name and `"Pending human review"` until explicitly confirmed by a team member; cross-model review completed; UI matches Stitch prototype; merged to `develop` with sprint review/retrospective notes.

| Risk ID | Risk Description | Impact | Mitigation & Trigger |
|---|---|---|---|
| `RISK-H01` | **Must gap of `+2.0 h` in S1+S2** (`53.0 h` Must vs `51.0 h` net) | Sprint 2 schedule pressure | Accepted by PO; 10% buffer kept intact; S3 Should stories absorb overrun; re-plan if S1 `human/classic > 0.45` or S2 Must exceeds net by `> 10%`. |

| Risk ID | Risk Description | Impact | Mitigation & Trigger |
|---|---|---|---|
| `RISK-H02` | **Sprint 3 Stretch stories ( `19.5 h` ) exceed S3 net ( `17.8 h` )** | Stretch items ( `LMS-014` , `016` , `021` , `018` .. `020` ) not reached | Explicit commitment line at `17.5 h <= 17.8 h` ; MVP ( `70.5 h` ) requires only the 4 committed Should stories above the line. |
| `RISK-H03` | **Weekly AI quota exhaustion** across `46` budgeted agent sessions | Interruption mid-story | Commit & push after every file/step; keep `docs/process/*_PROGRESS.md` updated; budget agent sessions per week. |
| `RISK-H04` | **Human-hour conversion ratio ( `~0.40x` ) is an assumption** | Velocity variance | Track actual human hours in `TASKS.md` during Sprint 1 and re-calibrate Sprint 2–3 estimates at Sprint 1 Review. |
| `RISK-H05` .. `H08` | **Midterm capacity drop ( `Nov 17–28` ), VM proxy Maven wrapper, HW01 open design items, dual ADUZEM upload** | Schedule/build/submission issues | Midterms prorated ( `3 h/wk` ); `tools/mvnw-vm.sh` isolated from standard `./mvnw` ; `LMS-008` spike in S1; mandatory dual-upload check. |

## 2.5 Potentially Shippable Increments (PSI) & MVP Definition

Authoritative source: `docs/02-inception/PSI_AND_MVP.md` .

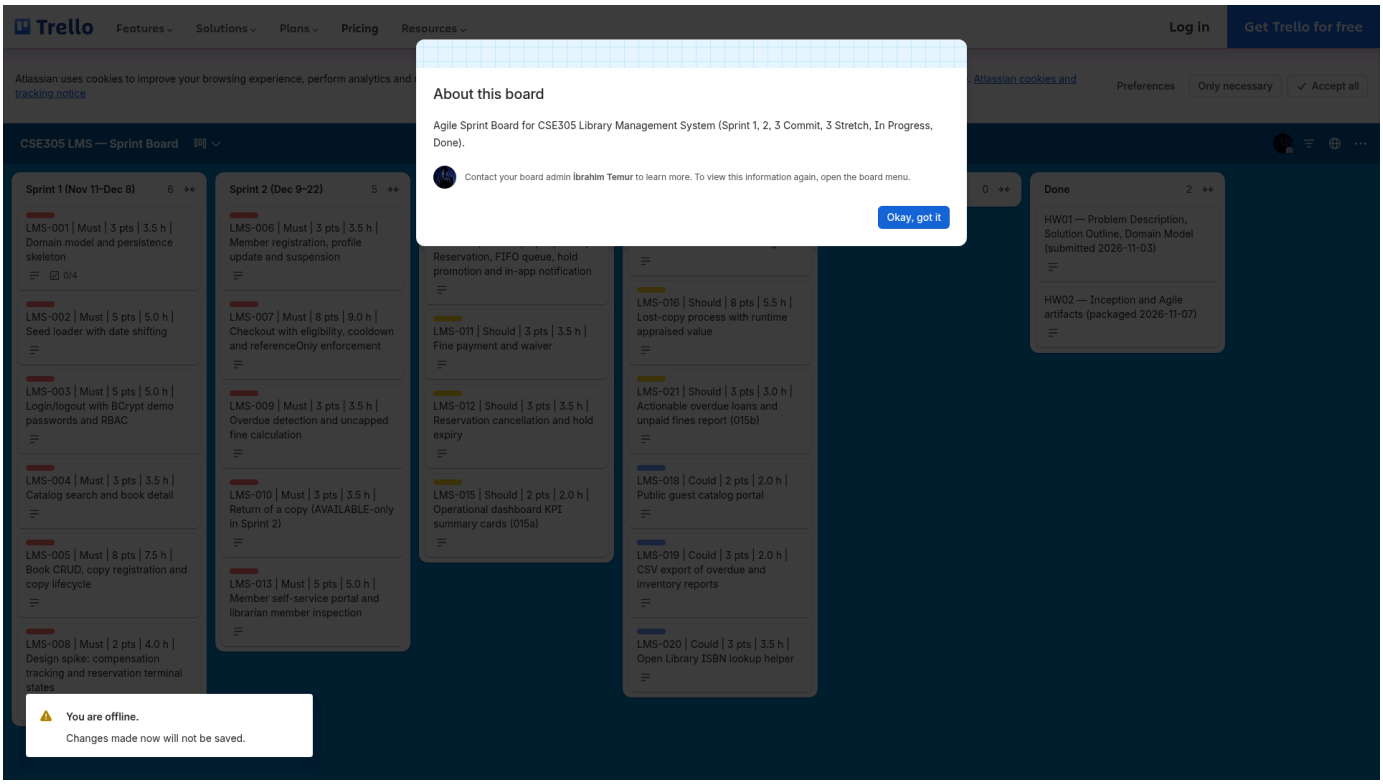
| Increment | Included Stories | Committed Effort | End-to-End Demonstrable Capability (PSI / MVP) |
|---|---|---|---|
| **Sprint 1 PSI ( `HW03` )** | `LMS-001` .. `LMS-005` , `LMS-008` ( `6` Must stories) | `26 pts` / `52` classic h / **`28.5` human h** | Application boots offline, loads seed v2 ( `782` books, `1,706` copies) with shifted dates ( `OPEN-04` ), authenticates Librarians and Members via BCrypt demo passwords ( `OPEN-05` ) and RBAC, supports catalog search/filtering and `referenceOnly` badges, enforces `BookCopyState` on Book/Copy CRUD, and documents the `LMS-008` ADR. |
| **Sprint 2 PSI ( `HW04` )** | + `LMS-006` , `LMS-007` , `LMS-009` , `LMS-010` , `LMS-013` ( `5` Must stories) | `22 pts` / `46` classic h / **`24.5` human h** *(Cumulative Must: `48 pts` / `53.0 h` )* | Complete circulation desk without holds: Librarian registers/suspends members, executes atomic checkouts ( `7` / `3` / `3` quotas, `30` -day period, `referenceOnly` & block checks, 30-day post-renewal cooldown) and returns ( `AVAILABLE` ), computes uncapped `0.25 TRY/day` fines, and Members inspect active/historical loans, cooldowns, and fines. |
| **Sprint 3 PSI — Committed MVP ( `HW05` )** | + `LMS-017` , `LMS-011` , `LMS-012` , `LMS-015` ( `4` Committed Should stories above the `17.5 h` line) | `16 pts` / `34` classic h / **`17.5` human h** ***(Total Committed MVP: `15` stories, `64 pts` , `132` classic h, `70.5` human h)*** | Adds title-level FIFO reservations, automatic `RESERVED` hold promotion on return with 3-day pickup window and in-app notification ( `LMS-017` ), desk fine `PAID` / `WAIVED` settlement ( `LMS-011` ), reservation cancellation and 3-day hold expiry reallocation ( `LMS-012` ), and operational dashboard KPI summary cards ( `LMS-015` ). |
| **Sprint 3 Stretch Slice *(Below Line, At Risk)*** | `LMS-014` , `LMS-016` , `LMS-021` , `LMS-018` .. `LMS-020` ( `3` Should + `3` Could stories) | `24 pts` / `41` classic h / **`19.5` human h** *(Full backlog if all stretch completed: `21` stories, `88 pts` , `90.0 h` )* | Self-service/desk renewal ( `LMS-014` ), lost-copy petition & `2x` runtime `appraisedValue` assessment ( `LMS-016` ), actionable overdue/unpaid report table ( `LMS-021` ), public guest catalog ( `LMS-018` ), CSV export ( `LMS-019` ), and live Open Library ISBN lookup helper ( `LMS-020` ). |

## 2.7 Sprint Tracking in Trello

Authoritative detail document: `docs/02-inception/TRELLO_BOARD.md` .

In accordance with the CSE305 HW02 Agile instructions (*"Use Trello to properly track and manage your sprints"*), sprint execution, work-in-progress visibility, and acceptance criteria verification are tracked using a dedicated, public Trello board.

- **Public Board URL:** https://trello.com/b/34K6O3Y1/cse305-lms-sprint-board
- **Board Architecture:** Structured into **6 lists** corresponding to the three delivery sprints and workflow states:
  1. `Sprint 1 (Nov 11–Dec 8)` (6 cards, 26 pts, 28.5 h — Must)
  2. `Sprint 2 (Dec 9–22)` (5 cards, 22 pts, 24.5 h — Must)
  3. `Sprint 3 Commit (Dec 23–Jan 5)` (4 cards, 16 pts, 17.5 h — Should)
  4. `Sprint 3 Stretch` (6 cards, 24 pts, 19.5 h — Should & Could)
  5. `In Progress` (active work items; max 2 cards per developer)
  6. `Done` (milestone deliverables `HW01` and `HW02` + verified increments satisfying the DoD)
- **Story Card Uniform Syntax:** `LMS-xxx | MoSCoW | N pts | H h | title` with color-coded native labels ( `Must` = Red, `Should` = Yellow, `Could` = Blue).
- **Backlog & Task Consistency:** All 21 user stories ( `88 pts` , `90.0 h` ) contain their Given/When/Then acceptance criteria and explicit story dependencies mirrored from `docs/02-inception/USER_STORIES.md` . Story `LMS-001` embeds the 4 granular technical tasks ( `T-001-1` .. `T-001-4` ) from `docs/02-inception/TASKS.md` as a native checklist.




## Revision History

| Version | Date | Summary |
|---|---|---|
| `1.0` | 2026-11-07 | First consolidated HW02 submission document: Part 1 consolidates the UP Inception Overview ( `VISION_AND_CONTEXT.md` , `PERSONAS.md` , `REQUIREMENTS.md` , `BUSINESS_RULES.md` , `USE_CASES.md` , `DOMAIN_GLOSSARY.md` ) with embedded System Context and Use Case diagrams; Part 2 presents the Product Owner-approved Agile Scrum Product Backlog ( `21` stories `LMS-001` .. `LMS-021` , `88` story points, `173.0` classic hours, `90.0` human hours, `105` tasks), 3-sprint capacity model, accepted `+2.0 h` Must gap governance ( `RISK-H01` ), Sprint 3 commitment line ( `17.5 h <= 17.8 h` net), Themes/Epics and Sprint Roadmap diagrams, dependency proof ( `OVERALL: PASS` ), and PSI/MVP definition ( `64 pts / 70.5 h` committed). |
| `1.1` | 2026-11-11 | Added Section 2.7 for Trello sprint board tracking (HW02 r4): incorporated public board URL ( `https://trello.com/b/34K6O3Y1/cse305-lms-sprint-board` ), 6-list workflow architecture, MoSCoW color labels, 21 story cards (88 pts, 90.0 h), LMS-001 technical task checklist, and cross-reference to `TRELLO_BOARD.md` . |